

DealLens technical assessment

Mizuho Derivatives Risk Solutions | Public source prototype

DealLens links three public transaction filings to structured evidence, source comparisons and illustrative financing sensitivities. The current prototype adds full numbered provisions, continuation pages and explicit reference context. It remains a submission candidate with incomplete normalized interpretation of complex clauses. Source excerpts are labeled partial; no human verification is claimed.

Implementation and evidence controls

Python ingestion preserves PDF hashes, page inventories, exhibit layers and page/character locators. A common 42-field catalog drives scalar extraction and bounded model proposals with exact citation checks. Explicit review decisions create new runs and regenerate dependent outputs; prior runs are preserved. SQLite retains request and review events. Reviewer identity is self-attested. The browser is read-only and loopback-bound.

Relevant executed agreements outrank summaries, but conflicting values and qualifiers are retained. Exact evidence matching proves provenance, not semantic correctness. Complex provisions return null normalized values and enter the review queue. All three PDFs match fresh publisher downloads. Organon and Uber filing dates are SEC-index-verified; Bio-Techne is explicitly PDF-metadata-reported. Omitted schedules and incorporated documents are not assumed available.

Document	Pages	Supported fields	Section excerpt fields
Bio-Techne	99	12 of 42	21
Organon	109	11 of 42	18
Uber and Delivery Hero	149	15 of 42	20

Supported counts exclude section excerpts and do not imply a resolved canonical comparison. Excerpts preserve contractual wording and references but do not establish a complete legal interpretation. All three sources use the same code. Common rules were revised after validation inspection, so the final results are adapted validation results, not pristine holdout accuracy.

Material source distinctions

Bio-Techne: March 25, 2027 outside date with specified automatic extensions to June 25 and September 25; further written agreement can extend it again. The debt-issuance assumptions in the assignment are synthetic, not disclosed financing terms. [1]

Organon: an elective extension can change the remedy obligations when Parent elects it. Award treatment also varies by grant year. Full sections preserve the grant-year cohorts, closing-time-condition exception and Parent-elected waiver for review. [2]

Uber: a German public takeover, not a US merger. The summary price is EUR 41.50; the agreement specifies at least EUR 41.50. A EUR 11.5bn financing floor is distinct from the disclosed EUR 14.2bn bridge commitment. The model preserves these distinctions. [3]

Financing and hedge analysis

All figures below use the assignment's synthetic USD 4bn, seven-year fixed-rate issue and USD 65,000 DV01 per USD 100mm. Portfolio DV01 is USD 2.6mm per basis point. Positive net incremental cost is worse for the issuer; positive hedge P&L is a receipt.

Benchmark and spread decomposition

Assumed bond coupon is Treasury plus issuer spread: $4.25\% + 1.00\% = 5.25\%$. The 4.40% swap rate implies a 15bp initial swap spread, which is not added again to the bond coupon. Debt cost sensitivity equals DV01 times the benchmark and issuer-spread moves. A payer swap follows the swap-rate move, leaving issuer-spread and swap-spread basis risk. Credit DV01 is assumed equal to benchmark DV01.

Scenario	Unhedged	Payer swap	Payer option	Contingent swap
Benchmark +25bp	65	0	12	8
Benchmark -25bp	-65	0	-53	8
Benchmark +50bp	130	0	12	8
Benchmark +25bp and credit +20bp	117	52	64	60
Failure and benchmark -25bp	0	65	12	8

Values are USD millions of incremental cost PV. Option premium is a synthetic 30bp of notional (USD 12mm); the contingent swap fee is a synthetic 20bp (USD 8mm), paid upfront in every outcome. Both assume matching hedge DV01. These are scenario assumptions, not market prices or executable terms.

Failure and timing

A conventional payer swap can require an unwind payment when the deal fails after rates fall. An ordinary payer option can still pay on a failure scenario with higher rates; a deal-contingent swap is assumed to cancel without settlement on failure, retaining the fee. The two strategies are therefore modeled separately. No transaction termination fee is automatically offset against hedge loss.

Delay scenarios use hypothetical closing on an explicitly stated extension date with separately labeled roll or renewal costs. They do not assert that extension conditions have been met. A +25bp rate move changes annual coupon cost by USD 10mm, distinct from the USD 65mm PV sensitivity; these measures are never summed. Outcome-weighted results depend on an explicitly synthetic joint rate/completion distribution.

Validation adaptations

Organon uses a synthetic USD 1bn fixed-rate refinancing slice because the baseline does not establish a supported issuance size and tenor. Uber uses a synthetic EUR 1bn slice plus separate FX and floating-bridge sensitivities. The bridge commitment is not assumed fully drawn, and its 364-day floating exposure does not use the seven-year bond DV01. A six-level disclosed bridge grid is now separate from assumed draw and EURIBOR. Step-up amounts, duration fees and funding fees are publicly redacted; no missing amount is inferred. [2, 3]

Validation and next steps

Verification evidence

The project passes 97 unit/control tests, 25 selected scalar/source fixtures and 18 section/qualifier retention fixtures. Tests cover financial identities, evidence integrity, qualifiers, unsupported QA, malformed model output, non-finite inputs, stale reviews and synthetic review regeneration. Browser checks confirmed full award sections, strict-mode abstention and the risk-map view. HTTP checks cover the supported answer and Uber price conflict. Agent-curated fixtures are not independent human ground truth; full-document semantic accuracy remains unmeasured.

Tests exposed ancillary-date contamination, a missed outside-date proviso, missing preceding retrieval context and failure to retrieve ISO-currency consideration. Shared rules and regression tests address these failures. Section tests retain continuation-page exceptions and references. Word boundaries prevent RSU/pursuant and cure/procure false matches. These tests do not measure full legal recall. No human decisions are fabricated.

Current scope and remaining work

Across 36 required transaction/question combinations, 12 answers are fully supported, 22 are partial and two remain review-required or conflicted. Complex partial answers now expose whole provisions and available references. They do not count as completed normalized legal extraction. Typed business-day expressions retain their contractual anchors without inventing calendar dates.

The section-aware continuation used no new provider calls and preserved earlier model proposals and budget records. Complete reconciliation of complex fee triggers, award cohorts, regulatory remedies and conditional deadlines remains outstanding. Redacted public financial terms cannot be recovered by guessing. Independent semantic precision and recall remain unmeasured; the candidate retains ownership of the submitted code, calculations and limitations.

Production and confidential information

For material non-public information, use firm-approved isolated infrastructure, authenticated deal-level access, private model endpoints, restricted egress, encrypted storage and approved retention terms. Reapply source permissions at retrieval and output. Keep credentials in a secrets service. Document text has no authority to execute tools or transmit data. Material corrections and distribution require attributed human approval; trading remains outside this application.

Replace illustrative prices with approved curves and volatility surfaces, add exact bridge cashflows and FX pricing, and reconcile against independent pricing tools. Add authenticated, signed review events, reproducible dependency locks, source-update monitoring and broader exception handling before operational deployment.

Primary source references

[1] Bio-Techne Form 8-K and merger agreement, supplied PDF: summary pp. 1-3; Sections 4.07, 5.06 and 7.01, PDF pp. 46, 60 and 71-72. [Original filing](#).

[2] Organon Form 8-K and merger agreement, supplied PDF: summary pp. 2-4; financing discussion pp. 59-60; Section 9.2 pp. 86-87; Section 9.5 pp. 88-89. [Original filing](#).

[3] Uber Form 8-K, BCA and bridge agreement, supplied PDF: summary pp. 2-4; BCA clauses 1.2, 1.4 and 1.5, pp. 13-15; bridge exhibit starts p. 44. [Original filing](#).